



IILM UNIVERSITY, GREATER NOIDA

GESTUREVOLUME: VOLUME CONTROL WITH HAND GESTURES

Internship Presentation

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Enrollment / Roll No.	CS-2341034
Program & Semester	B.Tech CSE (AI/ML) • Semester VII
Section	4CSE3
Organization	Infosys Springboard
Internship	Internship 7.0 (B1)
Duration	19 June 2026 – 14 August 2026

TOUCHLESS

Volume
Control



Webcam • OpenCV • MediaPipe
Flask • Real-Time Gesture Detection

Internship Overview

01



Organization

Infosys Springboard

Provided a structured virtual internship environment focused on practical, project-based learning.



Internship

Virtual Internship 7.0 (Batch 1)

19 June 2026 – 14 August 2026

A virtual, project-focused internship with hands-on learning.



Project

GestureVolume

A real-time computer-vision system for controlling computer volume through hand gestures.

What I worked on

01



Hand Tracking

Implemented real-time hand landmark detection using MediaPipe and OpenCV to capture hand movements from the webcam.

02



Gesture-Based Control

Mapped the distance between thumb and index finger to system volume, with smoothing and calibration for stable and accurate control.

03



Web Dashboard

Built a Flask-based web interface with live video stream, real-time volume status, detection controls, FPS monitoring and volume history.

Problem Statement

02

FROM GESTURES TO CONTROL



The challenge

Traditional computer-volume control requires physical interaction through a keyboard, mouse, touch interface or dedicated hardware. This can be inconvenient, especially in hands-free situations, and may not be accessible for everyone.

Not hands-free

Less convenient

Limited accessibility



The proposed solution

Use a webcam to detect hand landmarks, measure the thumb–index finger distance, map it to a volume percentage, and provide immediate visual feedback through a real-time dashboard — enabling touchless, intuitive and accessible volume control.

Touchless control

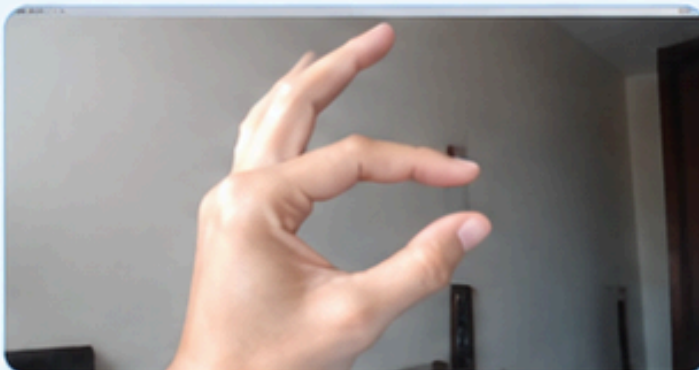
Real-time feedback

User-friendly

01

Input

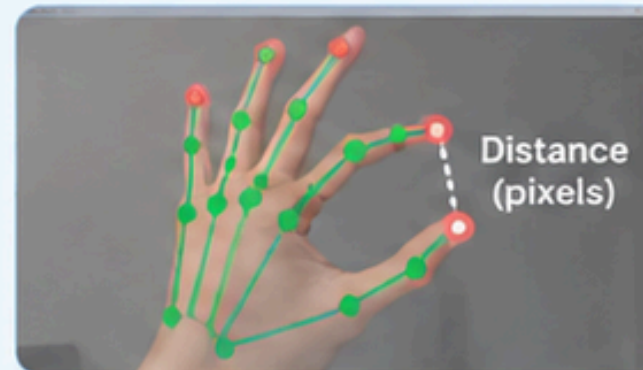
Live webcam frames



02

Processing

Hand landmarks + distance



03

Output

System volume + dashboard



Project Objectives

03

Key goals to design, implement and evaluate a real-time hand-gesture based volume control system.

01



Develop a real-time hand-gesture mechanism for system-volume control.

02



Capture and process live webcam frames using OpenCV.

03



Detect hand landmarks using MediaPipe Hands.

04



Map the thumb-index distance to a 0–100% volume range.

05



Apply smoothing to reduce unstable frame-to-frame changes.

06



Integrate the vision loop with a Flask-powered dashboard.

07



Provide manual minimum/maximum calibration.

08



Provide detection controls and performance feedback such as FPS.

Technologies & Tools

04

A combination of computer vision, web development and system control tools to build a real-time, interactive application.



Python

Core programming language for application logic, computer vision processing and system integration.



OpenCV

Webcam capture, image processing, drawing annotations and frame encoding for real-time video stream.



MediaPipe

Real-time hand landmark detection and tracking for gesture recognition.



Flask

Web backend to serve the dashboard, live video stream and status API.



HTML / CSS / JavaScript

Dashboard structure, styling and client-side interactivity for a smooth user experience.



Pycaw

Control system volume on Windows through the audio endpoint.



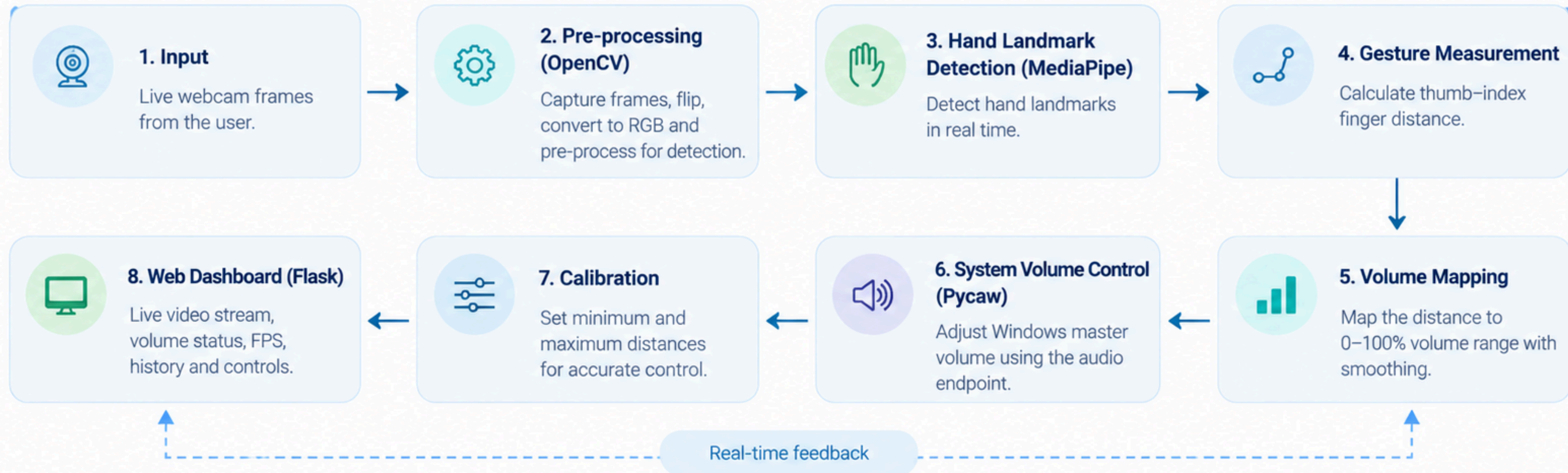
Development focus: Real-time computer vision + practical human-computer interaction.

System Architecture

05

An overview of the end-to-end pipeline showing how hand gestures are processed and converted into system volume control with a real-time web interface.

GestureVolume — System Architecture



Methodology / Working

05

Step-by-step workflow of how GestureVolume processes hand gestures and controls the system volume in real time.



Real-time hand gesture recognition → volume control → instant visual feedback

Key Features & Dashboard

07

An interactive dashboard for real-time gesture-based volume control.



Real-time Gesture Control

Adjust system volume by changing the distance between thumb and index finger.



Manual Calibration

Set minimum and maximum gesture distances for personalized control.



Smoothing

Rolling-average smoothing reduces rapid frame-to-frame volume changes.



Live Dashboard

View camera stream, volume, detection status and FPS in real time.



Volume History

Track recent volume changes with a live history graph.



Detection Controls

Pause/resume detection and switch dashboard theme.

Gesture Based Volume Control Web App

Control system volume using hand gestures

Camera Feed

Live Metrics

Distance	239 px
Volume	100%
Min Hands	1
FPS	7
Connection	ONLINE
Distance Range	HIGH

Calibration Status

Calibration not started

MIN Distance	20 px
MAX Distance	200 px

START CALIBRATIONSTOP CALIBRATIONRESETDEFAULT

Calibration Active

How to Calibrate

- Keep Hand Visible**
Ensure your hand is fully inside the camera frame.
- Start Close**
Bring thumb and index finger very close together.
- Move Apart Slowly**
Gradually increase the distance between fingers.
- Stop Calibration**
Click Stop Calibration once full range is covered.

Best Practices

- Use bright and even lighting
- Keep background simple
- Move fingers slowly for accuracy
- Use full finger range



Touchless control • Real-time feedback • Personalized calibration

Outcomes & Learning

08

Key results achieved and skills developed during the internship.

Project outcomes

- 1 Real-time webcam-based hand detection and tracking**
Implemented accurate hand landmark detection using MediaPipe for real-time processing.
- 2 Thumb-index distance mapped to continuous system-volume control**
Converted the distance between thumb and index finger into a smooth 0–100% volume range.
- 3 Rolling-average smoothing for stable volume behavior**
Applied temporal smoothing to reduce rapid frame-to-frame fluctuations.
- 4 Flask dashboard with live video, status, volume and FPS feedback**
Built a web interface to visualize the camera stream and real-time system status.
- 5 Pycaw-based Windows master-volume control**
Integrated Pycaw to adjust the system's master volume based on detected gestures.
- 6 Calibration and detection controls for adaptable use**
Added manual min/max calibration and controls to improve accuracy across different users and environments.

Skills strengthened

- | | |
|--|---|
|  Python programming
Core logic, integration and automation |  Computer vision
Image processing and real-time analysis |
|  MediaPipe hand tracking
Hand landmark detection and tracking |  Flask web development
Dashboard, live video and backend integration |
|  Frontend integration
UI structure and real-time display |  Real-time system design
Optimizing for low latency and smooth performance |
|  Debugging & testing
Error handling, testing and performance tuning |  Technical documentation
Project report, code comments and user guide |



From computer vision concepts to a working real-time application.

Internship Completion Certificate

Infosys Springboard • Internship 7.0 (B1) • 19 June 2026 – 14 August 2026



CERTIFICATE OF COMPLETION

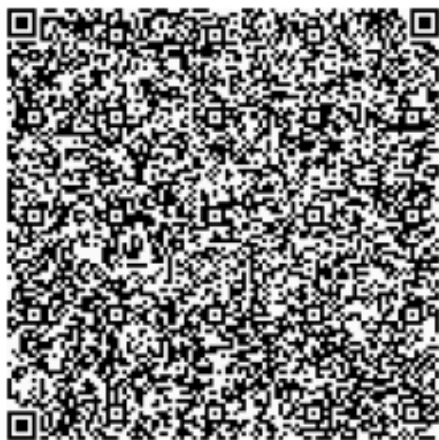
presented to

Shreya .

for actively participating and completing themandatory assignment related to

Internship 7.0 (B1) GestureVolume: Volume Control with Hand Gestures

conducted from June 19, 2026 to August 14, 2026



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To verify, scan the QR code at <https://verify.onwingspan.com>

Satheesha B.N.
Satheesha B. Nanjappa
Senior Vice President and Head
Education, Training and Assessment
Infosys Limited

that's all

THANK YOU

GestureVolume: Volume Control with Hand Gestures

A practical demonstration of computer vision, real-time processing and touchless human-computer interaction.



Gesture Detect



Adjust Volume



Real-time Processing



Smarter Interaction

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Infosys Springboard Internship 7.0 (B1)



*Small Gestures
Big Possibilities*